

Bacterial Glucarate and Galactarate Catabolism to 2-Oxoglutarate: A Review of a Reusable Aldarate-Oxidation Module

A concise review-style synthesis. Search date: 2026-08-13. Sources: primary literature and authoritative reviews retrieved via PubMed.

1. Executive summary

Bacteria that live on the diacid sugars **D-glucarate** and **D-galactarate (mucate)** funnel both substrates into a single shared intermediate, **5-dehydro-4-deoxy-D-glucarate** (equivalently 5-keto-4-deoxy-D-glucarate, "KDG," or 2-keto-3-deoxy-hexarate), using two **non-homologous entry dehydratases**. This convergence is the defining architectural feature of the module: glucarate is dehydrated by an enolase-superfamily Mg^{2+} enzyme (glucarate dehydratase, GudD/GlucD), while galactarate is dehydrated by one of several structurally and mechanistically distinct galactarate dehydratases (the *Escherichia coli* GarD fold, or enolase-superfamily paralogues such as TalrD/GalrD and GalrD-II). From the shared intermediate onward, two metabolic fates exist and they are frequently confused. The **route to 2-oxoglutarate**—the subject of this review—is the terminal segment of the **oxidative uronate pathway** found in organisms such as *Agrobacterium tumefaciens*, *Azospirillum brasilense*, and *Pseudomonas*: a **decarboxylating keto-deoxy-glucarate (KDG) dehydratase** converts the shared intermediate to **2,5-dioxopentanoate** (α -ketoglutarate semialdehyde, α -KGSA) plus CO_2 and H_2O , and an **ALDH-superfamily 2,5-dioxovalerate/ α -KGSA dehydrogenase** (NAD^+ - or $NADP^+$ -dependent) then oxidizes this aldehyde to **2-oxoglutarate**, feeding it directly into the TCA cycle.

The single most important boundary clarification is that the **canonical *E. coli* glucarate/galactarate pathway does not make 2-oxoglutarate**. In *E. coli*, the same shared KDG intermediate is cleaved by an **aldolase (GarL)** to pyruvate and tartronate semialdehyde, which is reduced (GarR) and phosphorylated (GarK) to 2-phosphoglycerate. The three-enzyme oxidative module and the *E. coli* aldolytic module thus share their entry chemistry and their

central intermediate but diverge completely at the branch point. Any review of "bacterial glucarate/galactarate catabolism to 2-oxoglutarate" is therefore a review of the oxidative terminal segment, not of the well-studied *E. coli* system that dominates the older literature.

A second unifying theme is **repeated, convergent evolution of the component enzymes**. Galactarate dehydratase activity has arisen at least four independent times in different protein folds; the terminal α -KGSA dehydrogenase is a promiscuously recruited node shared with hydroxyproline catabolism and non-phosphorylative Entner–Doudoroff sugar pathways; and the entry glucarate dehydratase sits within the enolase superfamily, the textbook paradigm for divergent evolution of enzyme function from a conserved partial reaction. The module is therefore best understood not as one conserved operon but as a **reusable, mix-and-match set of interchangeable catalytic parts** that different lineages have assembled independently to achieve the same net transformation: two six-carbon aldarc acids $\rightarrow$ one five-carbon TCA intermediate + CO_2 .

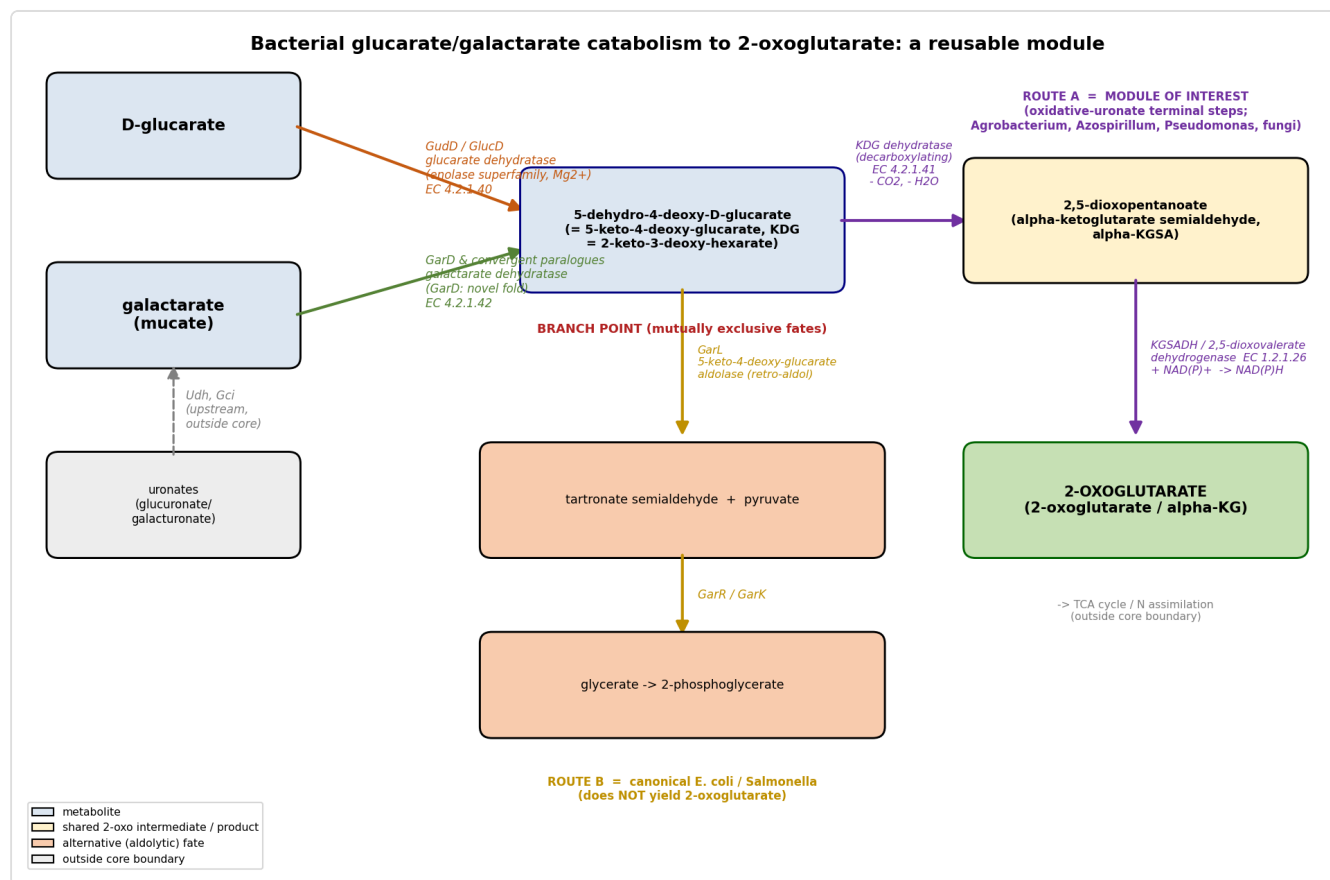


Figure 1. Schematic of the glucarate/galactarate → 2-oxoglutarate module. Two non-homologous entry dehydratases (GudD/GlucD for glucarate; GarD and its convergent paralogues for galactarate) converge on the shared intermediate 5-dehydro-4-deoxy-D-glucarate (KDG). The oxidative branch (right) uses a decarboxylating KDG dehydratase to make 2,5-dioxopentanoate (α -KGSA), which an α -KGSA/2,5-dioxovalerate dehydrogenase oxidizes to 2-oxoglutarate. The canonical *E. coli* branch (left) instead cleaves KDG with an aldolase (GarL) to pyruvate + tartronate semialdehyde, en route to 2-phosphoglycerate.

2. Definition and biological boundaries

2.1 What is inside the module

The core system comprises three catalytic steps and one branch decision:

- Alternative aldarate entry dehydration.** D-glucarate → KDG (glucarate dehydratase, GudD/GlucD); D-galactarate → KDG (galactarate dehydratase, GarD or paralogues). These are **alternative, mutually substitutable entry routes** that converge on a common product.

2. **Shared 2,5-dioxopentanoate formation.** $\text{KDG} \rightarrow 2,5\text{-dioxopentanoate } (\alpha\text{-KGSA}) + \text{CO}_2 + \text{H}_2\text{O}$, by a **decarboxylating dehydratase** (KDG dehydratase; a class-I aldolase/DHDPs-superfamily fold in the oxidative pathway).
3. **Terminal oxidation to 2-oxoglutarate.** $\alpha\text{-KGSA} + \text{NAD(P)}^+ + \text{H}_2\text{O} \rightarrow 2\text{-oxoglutarate} + \text{NAD(P)H}$, by a **2,5-dioxovalerate dehydrogenase** (α -KGSA dehydrogenase, KGSADH; EC 1.2.1.26).

2.2 What is outside the module (and often confused with it)

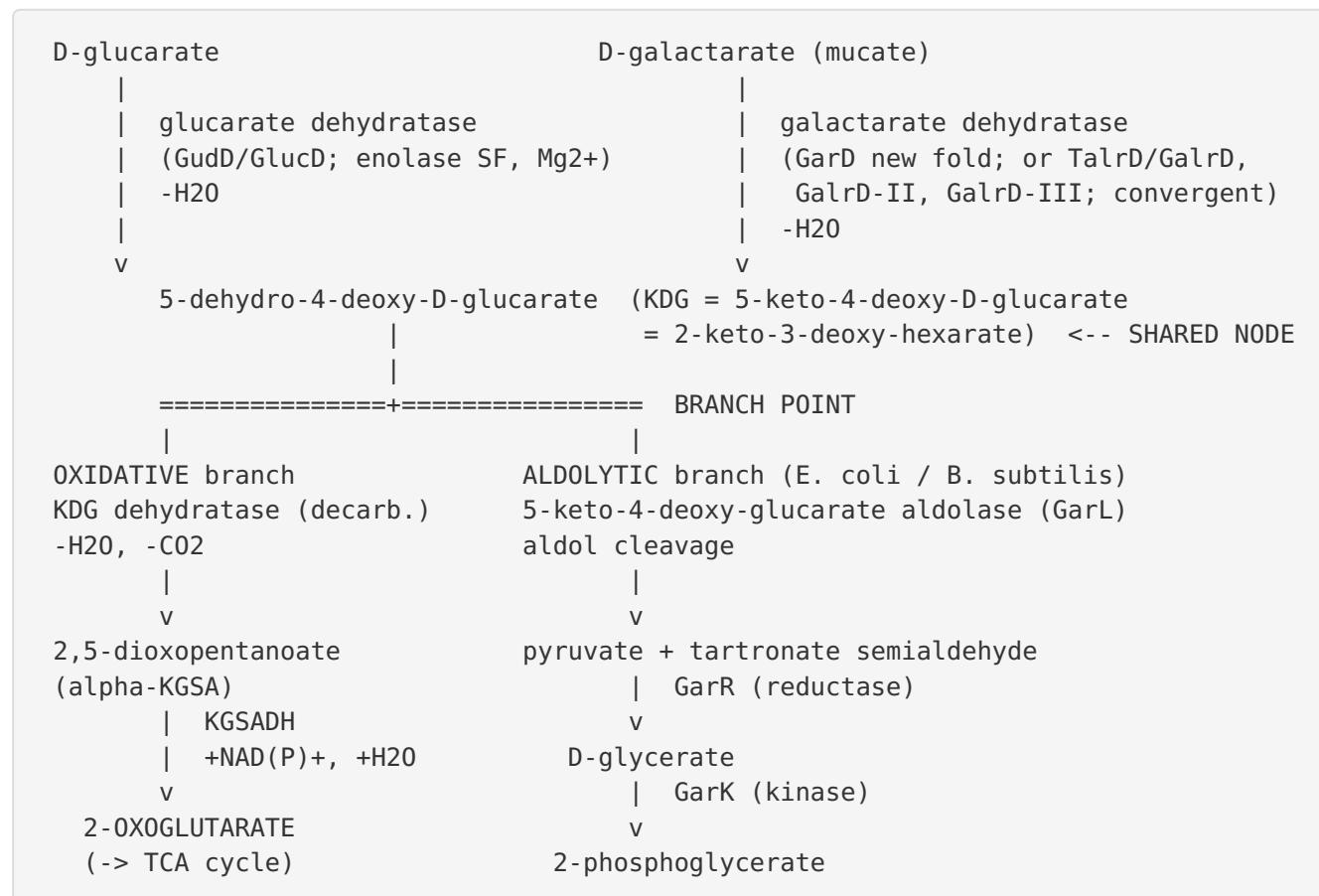
- **Transport and regulation.** Glucarate/galactarate uptake (e.g., the GudP/GarP permeases) and the transcriptional regulator (CdaR/SdaR/YaeG) coordinate the pathway but lie outside the catalytic core. They are treated here only as context (§5).
- **Upstream uronate oxidation.** In the oxidative uronate pathway, galactarate is produced from D-galacturonate by **uronate dehydrogenase (Udh)** and a **galactarolactone isomerase/cycloisomerase (Gli/Gci)** step. This chemistry supplies the module's substrate but is not part of the aldarate-to-2-oxoglutarate conversion.
- **The *E. coli* aldolytic fate.** The **GarL aldolase** $\rightarrow$ **GarR reductase** $\rightarrow$ **GarK kinase** branch that yields 2-phosphoglycerate and pyruvate is a *different terminal module* that shares the entry steps and the KDG intermediate. It is the single most important "look-alike" pathway and must be treated separately (Finding F002).
- **Downstream central metabolism.** Once 2-oxoglutarate is formed, its fate in the TCA cycle, in amino-acid biosynthesis, or as a signaling metabolite is outside scope.

2.3 Competing definitions in the literature

The phrase "glucarate/galactarate catabolism" is used in two incompatible senses. The **enteric (*E. coli*/*B. subtilis*)** sense ends in glycerate/2-phosphoglycerate via aldol cleavage; the **oxidative-uronate (*Agrobacterium*/*Azospirillum*/*Pseudomonas*)** sense ends in 2-oxoglutarate via decarboxylation and oxidation. Both are experimentally validated in their respective organisms; they are not contradictory but describe different organisms' wiring downstream of a shared intermediate. Reviews that mix the two—citing *E. coli* entry enzymes but an oxidative terminus, or vice versa—risk implying a pathway that no single organism runs end-to-end. This review treats the 2-oxoglutarate-yielding wiring as the defining boundary.

3. Mechanistic overview

3.1 The best current model, step by step



Obligatory steps. Along the 2-oxoglutarate route, all three catalytic steps are obligatory: entry dehydration, decarboxylative dehydration, and terminal oxidation. There is no known shortcut from an aldarate directly to 2-oxoglutarate.

Conditional / alternative steps. The *identity* of the entry dehydratase is conditional on the substrate (glucarate vs. galactarate) and on which enzyme family a given genome encodes. The *cofactor* of the terminal dehydrogenase is conditional on the isozyme (NAD^+ vs. NADP^+). The upstream production of galactarate from galacturonate (Udh + Gci) is accessory and organism-specific.

Accessory / branch-defining step. The decision between decarboxylative oxidation ($\rightarrow$ 2-oxoglutarate) and aldol cleavage ($\rightarrow$ glycerate) is the single accessory determinant that fixes pathway outcome. It is set by which enzyme acts on KDG: a DHDPS/class-I-aldolase-fold **decarboxylating dehydratase** versus a **lyase/aldolase (GarL)**.

3.2 Chemistry of each step

Step 1 — entry dehydration. Glucarate dehydratase is a Mg^{2+} -dependent enolase-superfamily enzyme in the mandelate-racemase (MR) subgroup. Its $(\beta/\alpha)_7\beta$ -barrel active site abstracts the C5 proton α to the carboxylate (Lys207 for L-idarate; the His339–Asp313 dyad for D-glucarate), generating a Mg^{2+} -stabilized enediolate; vinylogous β -elimination of the C4 hydroxyl then yields KDG. Crystallography places the essential Mg^{2+} ligands at Asp235, Glu266, and Asn289, with Asn341/His339 implicated as the general acid facilitating leaving-group departure. Galactarate dehydration reaches the *same* KDG product, but by enzymes with different active-site acid/base machinery (§4).

Step 2 — decarboxylative dehydration. In the oxidative pathway (characterized in *Agrobacterium tumefaciens*), KDG dehydratase catalyzes a **decarboxylating hydro-lyase** reaction: net loss of water and CO_2 to give α -ketoglutarate semialdehyde (2,5-dioxopentanoate). Structurally this enzyme belongs to a class-I aldolase (DHDPs/NAL) fold—distinct from the enolase-superfamily entry enzyme—underscoring that the module is assembled from unrelated scaffolds.

Step 3 — terminal oxidation. The α -KGSA/2,5-dioxovalerate dehydrogenase (KGSADH) is a member of the **aldehyde dehydrogenase (ALDH) superfamily**. It oxidizes the C1 aldehyde of α -KGSA to a carboxylate using NAD^+ or $NADP^+$, producing 2-oxoglutarate. The same enzymatic logic (oxidation of α -KGSA to 2-oxoglutarate) is reused verbatim in hydroxyproline catabolism and in non-phosphorylative Entner–Doudoroff pentose pathways (§4, §5).

4. Major molecular players and active assemblies

4.1 Finding F001 — two non-homologous entry dehydratases converge on one intermediate

In *E. coli* the glucarate/galactarate genes are organized in three transcriptional units (*garD*; *garPLRK*; *gudPD*), encoding **D-glucarate dehydratase (GlucD/GudD)**, an enolase-superfamily Mg^{2+} enzyme, and **galactarate dehydratase (GarD)**, a distinct fold. Both entry dehydratases produce the common intermediate 5-keto-4-deoxy-D-glucarate (= 2-keto-3-deoxy-hexarate = 5-dehydro-4-deoxy-D-glucarate). The mechanism of GlucD is established crystallographically:

Lys207 and the His339–Asp313 dyad abstract the C5 proton, and Mg^{2+} is ligated by Asp235/Glu266/Asn289 (PMID: 10769114). The convergence of two chemically unrelated dehydratases on a single product is the architectural crux of the module (PMID: 9772162).

4.2 Finding F002 — the 2-oxoglutarate route is the oxidative terminus, not the *E. coli* aldolase route

This is the pivotal boundary result. In *E. coli*, KDG is cleaved by **5-keto-4-deoxy-D-glucarate aldolase (GarL)** to pyruvate + tartronate semialdehyde, reduced by GarR and phosphorylated by GarK to 2-phosphoglycerate—it **does not yield 2-oxoglutarate** (PMID: 9772162). The route to 2-oxoglutarate is instead the oxidative-uronate terminal module: a **decarboxylating KDG dehydratase** (*Agrobacterium tumefaciens*; class-I aldolase fold) converts KDG to α -ketoglutarate semialdehyde (α -KGSA = 2,5-dioxopentanoate) + CO_2 + H_2O (PMID: 25454257), which is then oxidized by α -KGSA dehydrogenase (KGSADH; EC 1.2.1.26) to 2-oxoglutarate. The two branches share entry chemistry and the KDG node but are otherwise fully distinct.

4.3 Finding F003 — galactarate dehydratase is a convergently evolved activity

At least four structurally/mechanistically distinct galactarate dehydratases are documented:

Enzyme (family)	Organism	Fold / superfamily	Distinguishing catalytic feature
GarD	<i>E. coli</i>	New three-domain fold (not the enolase barrel)	Linked to fitness after antibiotic treatment (PMID: 31811683)
TalrD/GalrD (L-talarate/ galactarate dehydratase)	<i>Salmonella typhimurium</i>	Enolase SF, MR subgroup	Lys197; His328–Asp301 dyad; dehydrates L-talarate ($k_{cat} \approx 2.1 \text{ s}^{-1}$) and galactarate (PMID: 17649980)
GalrD-II	<i>Oceanobacillus theyensis</i>	Enolase SF	Regiochemically distinct Tyr164–Arg162 base; galactarate $k_{cat} \approx 6.8 \text{ s}^{-1}$, $K_m \approx 620 \text{ }\mu\text{M}$ (PMID: 19883118)
Galactarate dehydratase III (A9CG74)	<i>Agrobacterium tumefaciens</i> C58	Enolase SF, MR subgroup	Distinct substrate handling; genome-proximal decarboxylating dehydratase supplies α -KGSA (PMID: 24926996)

These enzymes achieve the same net dehydration by different active-site chemistry—a clear case of **convergent (and, within the enolase superfamily, "pseudoconvergent") evolution**. The full-length *E. coli* GarD structure revealed a **new protein fold**, distinct from the enolase-superfamily galactarate dehydratases ([PMID: 31811683](#)). For understanding the ancestral entry role, the enolase-superfamily MR-subgroup members (glucarate dehydratase and TalrD/GalrD) are the best representatives, because they retain the conserved superfamily partial reaction; the *E. coli* GarD "new fold" is a lineage-specific solution.

4.4 Finding F004 — the terminal dehydrogenase is a shared convergence node

In *Azospirillum brasilense*, distinct KGSADH isozymes converge on the same reaction (α -KGSA $\rightarrow$ 2-oxoglutarate): a **D-glucarate/D-galactarate-inducible NAD⁺-dependent KGSADH-II** and a **hydroxy-L-proline-inducible NADP⁺-dependent KGSADH-III**, alongside the L-arabinose-pathway KGSADH-I ([PMID: 17202142](#)). They share high specificity for α -KGSA but differ in coenzyme preference and are only poorly related within the ALDH superfamily—**molecular and metabolic convergent evolution**. The same aldehyde-oxidation logic recurs in

hydroxyproline catabolism and in non-phosphorylative Entner–Doudoroff sugar pathways, and archaeal ALDH paralogues (e.g., in *Sulfolobus solfataricus*) show physiologically significant activity toward α -KGSA, reinforcing the deep reuse of this step.

4.5 Finding F005 — co-regulation and colonization fitness

In *E. coli* the *gar* (*garD*; *garPLRK*) and *gud* (*gudPD*) operons are coordinately induced by D-galactarate, D-glucarate, and D-glycerate via a single common regulator (**CdaR/SdaR/YaeG**), which is autogenously regulated (PMID: 10762278). In *Bacillus subtilis* a single operon (*ycbCDEFGHJ*, including both glucarate and galactarate dehydratases) is induced by either D-glucarate or D-galactarate (PMID: 12044674). Galactarate/glucarate catabolism (GarD) increases the colonization fitness of intestinal pathogens in antibiotic-treated mice and promotes bacterial survival during stress; the pathway is widespread in bacteria but absent in humans (PMID: 31811683). Although transport and regulation are outside the catalytic core, this co-regulation explains why the two entry routes are functionally interchangeable in vivo—both inducers switch on both dehydratases.

4.6 Finding F006 — deepest origin lies in the enolase superfamily

The entry step's deepest plausible origin is the **enolase superfamily**, the paradigm for divergent evolution of enzyme function. All members share a conserved partial reaction—Mg²⁺-assisted abstraction of the α -proton of a carboxylate to form an enolate intermediate—but catalyze different overall reactions (racemization, dehydration, cycloisomerization, β -elimination) (PMID: 8987982; PMID: 22069326). Glucarate dehydratase belongs to the mandelate-racemase subgroup. Laboratory evolution shows that a monofunctional progenitor can acquire a "new" superfamily reaction by a single base change (*E. coli* AEE D297G gaining o-succinylbenzoate synthase activity), then be optimized by additional substitutions (I19F) (PMID: 18020459). This provides a mechanistic model for how substrate-specific acid-sugar dehydratases—including galactarate dehydratases—recur independently across lineages.

5. Evolutionary and cell-biological variation

5.1 Variation across lineages

- **Enteric bacteria** (*E. coli*, *Salmonella*, *B. subtilis*). Use enolase-superfamily glucarate dehydratase for entry, but terminate the shared intermediate by **aldol cleavage** (GarL) →

glycerate/2-phosphoglycerate. These organisms do **not** convert glucarate/galactarate to 2-oxoglutarate. *E. coli* nonetheless serves as the reference for the entry chemistry and for pathway regulation.

- **Soil/plant-associated α -proteobacteria and others (*Agrobacterium*, *Azospirillum*, *Pseudomonas*).** Terminate the shared intermediate by **decarboxylative dehydration + oxidation** $\rightarrow$ 2-oxoglutarate. This is the oxidative uronate pathway; galactarate here often arises upstream from D-galacturonate via uronate dehydrogenase and galactarolactone cycloisomerase.
- **Archaea.** Non-phosphorylative Entner–Doudoroff pathways in *Sulfolobus* generate α -KGSADH-type semialdehydes that are oxidized by ALDH-superfamily enzymes; some *Sulfolobus* paralogues have significant α -KGSADH activity, demonstrating that the terminal oxidation logic predates and extends beyond the bacterial aldarate module.

5.2 Physiological-state variation

The module is **inducible**, not constitutive: both dehydratases and (in *Azospirillum*) the specific KGSADH isozyme are induced by glucarate/galactarate. Cofactor usage of the terminal step is a physiological variable (NAD^+ vs. NADP^+), which likely tunes the step to a cell's redox/biosynthetic demands. Because bacteria lack the compartmentalization of eukaryotes, there is no organelle localization to consider; "compartmentalization" here reduces to substrate-specific and inducer-specific enzyme expression.

5.3 Alternative routes to the same outcome

Multiple molecular means achieve the same net result: - **Entry:** four independent galactarate dehydratase families + the enolase-superfamily glucarate dehydratase. - **Terminal oxidation:** multiple, poorly related KGSADH isozymes with different cofactor preferences, shared with hydroxyproline and pentose catabolism.

This redundancy is the practical meaning of calling the system a "**reusable module**."

6. Constraints, dependencies, and failure modes

6.1 Obligatory ordering

The three catalytic steps are strictly sequential and chemically committed: dehydration must precede decarboxylative dehydration, which must precede terminal oxidation. Each step's product is the next step's obligatory substrate; there is no bypass that reaches 2-oxoglutarate from an aldarate without passing through KDG and then α -KGSA.

6.2 The branch point is mutually exclusive

At the shared KDG node, **decarboxylative oxidation and aldol cleavage are mutually exclusive fates**. An organism's outcome is fixed by which enzyme it encodes to act on KDG. The evidence that rules out the aldolytic path yielding 2-oxoglutarate is direct: in *E. coli* the products are pyruvate + tartronate semialdehyde $\rightarrow$ glycerate $\rightarrow$ 2-phosphoglycerate, with no oxidative decarboxylation to α -KGSA ([PMID: 9772162](#)). Conversely, the oxidative branch's decarboxylating dehydratase and KGSADH are required to reach 2-oxoglutarate ([PMID: 25454257](#); [PMID: 17202142](#)).

6.3 Substrate specificity as a constraint

Entry is substrate-specific: glucarate dehydratase and galactarate dehydratase are not interchangeable at the enzyme level even though their products are identical. A genome lacking a galactarate dehydratase cannot use galactarate even if it has glucarate dehydratase, and vice versa—hence the value of co-regulating both under a single inducer set.

6.4 Failure modes

- **Loss of a single entry dehydratase** restricts substrate range but leaves the other route intact.
- **Loss of the branch enzyme (decarboxylating dehydratase)** redirects flux or blocks 2-oxoglutarate production.
- **Cofactor mismatch** (e.g., an NADP⁺-only KGSADH under NADP-limited conditions) could bottleneck terminal oxidation.
- Engineering studies exploit these nodes deliberately: deleting *garK* in *E. coli* reroutes flux to accumulate D-glyceric acid ([PMID: 33057913](#)), confirming the branch's control over pathway output.

7. Controversies and open questions

1. **Pathway conflation.** The most consequential "controversy" is definitional rather than empirical: literature and databases frequently splice *E. coli* entry enzymes onto an oxidative terminus, implying a 2-oxoglutarate-producing pathway that no single reference organism demonstrably runs end-to-end from glucarate. Cross-organism synthesis should be explicit that entry (enolase-SF, best characterized in *E. coli*/*Pseudomonas*) and terminus (oxidative, best characterized in *Agrobacterium*/*Azospirillum*) are documented in different organisms.
2. **How many galactarate dehydratase families exist, and which is ancestral?** Four are documented, spanning a novel fold (GarD) and three enolase-superfamily variants with different acid/base machinery. Whether these represent independent recruitments or repeated divergence from a promiscuous enolase-superfamily progenitor is not fully resolved; the enolase-superfamily members are the better guide to the ancestral entry chemistry, while GarD's distinct fold is a lineage-specific replacement.
3. **Which decarboxylating-dehydratase / KGSADH pairing operates in a given genome?** The oxidative branch enzymes are best characterized in a handful of organisms. The distribution, cofactor preference, and regulation of KGSADH isozymes across bacteria are incompletely mapped, and the assignment of specific genes to the 2-oxoglutarate-producing step in many genomes remains inferential.
4. **Physiological driver of NAD⁺ vs. NADP⁺ preference** at the terminal step is unresolved—whether it reflects redox balancing, biosynthetic NADPH supply, or historical contingency of enzyme recruitment.
5. **In vivo flux control** at the KDG branch point—how organisms that encode both aldolytic and oxidative capabilities (if any) partition flux—has not been quantitatively dissected.

8. Limitations and knowledge gaps

- **Organism heterogeneity.** The complete 2-oxoglutarate route is assembled from data across several organisms (entry in *E. coli*/*Pseudomonas*/*Salmonella*; terminus in *Agrobacterium*/*Azospirillum*). No single hyper-characterized organism anchors the entire module end-to-end, so cross-organism inference carries some risk.

- **Gene-to-function assignment.** For the oxidative branch, decarboxylating-dehydratase and KGSADH gene assignments are firm in a few strains but inferential in most genomes.
- **Quantitative flux and regulation** of the oxidative branch (as opposed to the well-studied *E. coli* aldolytic branch) are sparsely characterized.
- **This review is literature-derived**, synthesizing 37 primary papers and reviews; it includes no new experimental data.

9. Proposed follow-up experiments/actions

1. **End-to-end reconstitution.** Reconstitute the full glucarate→2-oxoglutarate route in vitro (or in a single heterologous host) using purified glucarate dehydratase, a decarboxylating KDG dehydratase, and a KGSADH, to confirm stoichiometry (2 H₂O + CO₂ lost; NAD(P)H produced) without organismal conflation.
 2. **Comparative genomics of branch-point enzymes.** Systematically map which genomes encode the decarboxylating dehydratase (oxidative) vs. the GarL aldolase (aldolytic), to predict pathway outcome from sequence and test whether any organism encodes both.
 3. **KGSADH cofactor determinants.** Solve structures and perform mutagenesis on NAD⁺- vs. NADP⁺-preferring KGSADH isozymes to define the cofactor-specificity switch and test whether it is engineerable.
 4. **Ancestral-state reconstruction** of galactarate dehydratase across the enolase superfamily and the GarD fold, to test independent-recruitment vs. repeated-divergence hypotheses and identify the best ancestral representative.
 5. **In vivo ¹³C flux analysis** in an oxidative-pathway organism to quantify flux from labeled glucarate/galactarate into 2-oxoglutarate and the TCA cycle, and to test branch-point partitioning under different physiological states.
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10. Key references

- *Characterization of the (D)-glucarate/galactarate catabolic pathway in Escherichia coli.* [PMID: 9772162](#) — defines the entry dehydratases, the shared KDG intermediate, and the aldolytic (non-2-OG) *E. coli* fate (F001, F002).

- *Crystallographic and mutagenesis studies of D-glucarate dehydratase from E. coli.* [PMID: 10769114](#) — GlucD active-site chemistry (Mg^{2+} ligands Asp235/Glu266/Asn289; Lys207; His339) (F001).
- *Identification of the general acid catalyst in E. coli D-glucarate dehydratase.* [PMID: 11513584](#) — mechanistic detail of the vinylogous elimination forming KDG.
- *Structure and function of a decarboxylating A. tumefaciens keto-deoxy-D-galactarate dehydratase.* [PMID: 25454257](#) — the decarboxylating dehydratase producing α -KGSA (2,5-dioxopentanoate) (F002).
- *Galactarate dehydratase III from A. tumefaciens C58.* [PMID: 24926996](#) — a further galactarate dehydratase and a genome-proximal decarboxylating dehydratase to α -KGSA (F003).
- *Structure of galactarate dehydratase (GarD), a new fold in bacterial fitness after antibiotics.* [PMID: 31811683](#) — E. coli GarD is a novel fold and a colonization-fitness factor (F003, F005).
- *L-talarate/galactarate dehydratase (TalrD/GalrD) from S. typhimurium.* [PMID: 17649980](#) — an enolase-superfamily galactarate dehydratase (F003).
- *Regiochemically distinct galactarate dehydratase (GalrD-II) from O. iheyensis.* [PMID: 19883118](#) — yet another convergent galactarate dehydratase family (F003).
- *α -KGSA dehydrogenase isozymes in D-glucarate/D-galactarate and hydroxy-L-proline metabolism (Azospirillum brasilense).* [PMID: 17202142](#) — convergent $\text{NAD}^+/\text{NADP}^+$ KGSADH isozymes; the terminal oxidation node (F004).
- *A common regulator for gar/gud/glycerate operons in E. coli.* [PMID: 10762278](#) — CdaR-type co-regulation by glucarate/galactarate/glycerate (F005).
- *B. subtilis D-glucarate/galactarate operon ycbCDEFGHJ.* [PMID: 12044674](#) — single operon co-induced by either aldarate (F005).
- *The enolase superfamily: abstraction of the α -protons of carboxylic acids.* [PMID: 8987982](#) — the conserved ancestral partial reaction (F006).
- *Divergent evolution in the enolase superfamily.* [PMID: 22069326](#) — the superfamily as the divergent-evolution paradigm (F006).
- *Stepwise in vitro evolution of a "new" reaction in the enolase superfamily.* [PMID: 18020459](#) — few mutations create new dehydratase-type activities (F006).
- *Catabolism of hexuronides, hexuronates, aldonates, and aldarates.* [PMID: 26443361](#) — authoritative review situating glucarate/galactarate catabolism and the E. coli end-products (pyruvate and 2-phosphoglycerate).

- *Production of D-glyceric acid from D-galacturonate in engineered E. coli.* [PMID: 33057913](#) — engineering evidence for branch-point flux control (garK deletion).
- *Structure/function of a novel keto-deoxy-D-galactarate (KDG) dehydratase from A. tumefaciens.* [PMID: 24419616](#) — structural characterization of the oxidative-branch decarboxylating enzyme.
- *Unraveling the function of ALDH paralogs from Sulfolobus solfataricus.* [PMID: 23296511](#) — archaeal ALDHs with α -KGSA activity, supporting deep reuse of the terminal step.

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